ESSENTIAL IMAGE CONTAINMENT FROM POINT-SHEAF AGREEMENT

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Adjoint existence. lemma-always-right-adjoints L26486 (carded this batch). F and G have adjoints.
- Right adjoint characterization. derived-lemma-right-adjoint (cross-chapter, named). Identifies the essential image with its double orthogonal.
- Orthogonal detection. lemma-orthogonal-point-sheaf L26488 (carded this batch). Vanishing against point sheaves kills the adjoint.

Source reference. Original reference 0CQS: focused-seeds/geometry/S9-eqv-two-functors-pre/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYN: sources/stacks-equiv/pinned-original-source.tex, source lines 1098–1098. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G03: sources/stacks-equiv/pinned-original-source.tex, source lines 2795–2795. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X, Y be proper schemes over k. Assume X is regular. Let $F, G : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$ be k-linear exact functors such that*

(1) *$F(\mathcal{F}) \cong G(\mathcal{F})$ for any coherent $\mathcal{O}_X$-module $\mathcal{F}$ with $\dim(Supp(\mathcal{F})) = 0$,*
(2) *F is fully faithful.*

Then the essential image of G is contained in the essential image of F.

Proof. Recall that F and G have both adjoints, see Lemma 0FYN. In particular the essential image $\mathcal{A} \subset D_{perf}(\mathcal{O}_Y)$ of F satisfies the equivalent conditions of Derived Categories, Lemma 0CQS. We claim that G factors through $\mathcal{A}$. Since $\mathcal{A} = {}^{\perp}(\mathcal{A}^{\perp})$ by Derived Categories, Lemma 0CQS it suffices to show that $\mathrm{Hom}_Y(G(M), N) = 0$ for all M in $D_{perf}(\mathcal{O}_X)$ and $N \in \mathcal{A}^{\perp}$. We have

$$\mathrm{Hom}_Y(G(M), N) = \mathrm{Hom}_X(M, G_r(N))$$

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where G_r is the right adjoint to G. Thus it suffices to prove that $G_r(N) = 0$. Since $G(\mathcal{F}) \cong F(\mathcal{F})$ for $\mathcal{F}$ as in (1) we see that

$$\mathrm{Hom}_X(\mathcal{F}, G_r(N)) = \mathrm{Hom}_Y(G(\mathcal{F}), N) = \mathrm{Hom}_Y(F(\mathcal{F}), N) = 0$$

as N is in the right orthogonal to the essential image $\mathcal{A}$ of F. Of course, the same vanishing holds for $\mathrm{Hom}_X(\mathcal{F}, G_r(N)[i])$ for any $i \in \mathbf{Z}$. Thus $G_r(N) = 0$ by Lemma 0G03 and we win. $\square$